

Technical Architecture Overview

Morris Law Kernel v3.5 + IP Protection System · Kevin Morris · July 2026

A concise technical summary of the Morris Law Kernel v3.5 + IP Protection System as implemented today. Suitable for sharing with technical evaluators at potential licensees and partners.

System Layers

Layer	Implementation	Status
DSP / Mastering	Morris Law Kernel v3.5 (Python + Numba JIT)	Production-ready
Watermarking	Robust hybrid embedding (spread-spectrum + echo hiding), TypeScript	Production-ready
Provenance	Split-key cryptographic record: content hash + perceptual hash + digital signature	Production-ready
Verification	FastAPI service + live public verification endpoint	Live
Frontend	Streamlit application (mastering, stem isolation, lyric pre-flight, verification)	Live
Anchoring	Blockchain / trusted timestamping module	Optional — roadmap

Mastering Engine (Morris Law Kernel v3.5)

- **Numba JIT compiled** core loops with parallel execution — the hot path runs as native machine code, not interpreted Python.
- **Stereo-linked adaptive compression** with program-dependent ratio; neutral at zero intensity (verified transparent, correlation > 0.9999).
- **Adaptive bass sidechain**: low-frequency energy detection that engages only on bass-heavy program material.
- **Auto-threshold**: tracks program level so one preset works across quiet and loud sources.
- **RBJ-accurate shelving EQ** (true biquad shelves, impulse-response verified flat passband) with Nyquist-safe clamping down to low sample rates.
- **BS.1770 loudness normalization** to target LUFS (measured accuracy within ±0.5 LU), with a transparent-gain ceiling — no clipping stage.
- **IntelligentMultiBandIsolator**: 5-band Linkwitz-style splitter with per-stem gating for drums / bass / vocals / air.
- **Honest failure modes**: silence is never amplified into fake loudness; unsupported inputs raise clear errors instead of corrupting output.

Measured Performance

- Approximately **109x realtime** mastering throughput measured on commodity CPU hardware (no GPU required).
- Full test suite covers JIT-vs-reference parity, determinism, loudness accuracy, low-sample-rate stability, and edge cases (empty, short, multichannel files).

IP Protection Chain

1. Audio is mastered by the kernel.
2. A provenance record is created: SHA-256 content hash, perceptual hash, creator identity, brand, timestamp.
3. The record is bound with a digital signature and embedded via robust hybrid watermarking designed to survive re-encoding.
4. **Split-key design**: the embedded payload alone is not enough to forge a certificate — the verification secret lives server-side only. A file can be verified by anyone; a valid certificate can be issued by no one but the server.
5. Verification is available through a live public API endpoint that returns a signed verdict report suitable for supporting copyright registration and disputes.

Integration Surfaces

- **Python API** — direct kernel integration for DAWs and pipelines.
- **REST verification API** — language-agnostic; any platform can verify files server-side.
- **Streamlit reference frontend** — demonstrates the full flow end to end.

Roadmap

- Blockchain / trusted-timestamp anchoring as an optional immutability layer.
- Native DAW plugin SDK.
- Expanded verification report formats for platform partners.